Agents in Version Control

A plain primer on what an agent runtime has to get right, and how Verlet does it

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1. The thing that breaks

Everyone wants an agent that gets better at its job over time. Today you can have one of two things. A personal agent that is allowed to change itself, which breaks often and needs constant repair. Or a hosted agent that never changes, which is safe for exactly that reason.

The breakage does not come from models being bad at improving themselves. It comes from where the agent keeps its state. The agent edits its own tools, prompts, or memory in place. Nothing records what changed or why. There is no undo. When something goes wrong, the only repair is to wipe the agent and start over. An agent you have to wipe never really had a history.

Software teams solved this problem for code a long time ago. Nobody lets a junior developer, or a bot, edit a production codebase without version control. An agent that edits itself needs the same protection. Two things go under version control, not one:

- The agent's **definition**: its models, tools, permissions, prompts, and procedures. This is the part that works like source code.
- The agent's **experience**: every run, every model output, every decision. This is the part that explains why a change was made.

Each change to the definition should point at the experience that caused it.

There is one place where the comparison to git does not hold. In git, a person decides when to commit. An agent cannot be given that choice. The agent is the one editing itself, and an agent that edits its own memory cannot be trusted to also write down that it did. So the runtime has to be built so that every action is automatically a commit. There must be no way to act without leaving a record.

Once that holds, self-modification stops being frightening. Every change the agent makes to itself is a commit with a reason attached. You can diff it, review it, and revert it. Trying a change is a branch. A bad change is a revert, and the agent keeps running.

This primer explains what a runtime has to get right for that to be true, and shows how Verlet, an experimental open-source runtime, does it. It is written for an engineer or product owner who is choosing what to build agents on, or who needs to check an agent product's claims. It uses very little special vocabulary. When a term is needed, it is introduced as the answer to a problem you have already seen.

> **If a section does not make sense, ask an agent.** Each section ends with a prompt. Clone the runtime (`git clone https://github.com/emotionscientific/verlet-kernel`), open Claude Code or any coding agent in that directory, and paste the prompt. The agent answers from the code, not from this document, so you can check what it says. Boxes marked **Lineage** are optional. They say where an idea came from in older computer science and where this design departs from it. Skip them on a first read.

One example agent runs through the whole document. It is a customer support agent. It reads incoming tickets, talks to customers, and can issue refunds through a payment API. Every piece of it exists in production somewhere today. Because it moves money, every question that matters comes up naturally: what happened, why, can we rerun it, and who allowed it.

Discussion

The version-control framing makes one claim and avoids another.

The claim: an agent has two kinds of state, and they belong in one history. Today they live in different places. A config file says what the agent is. A log says what it did. Nothing records that a line in the config exists because of an episode in the log. Putting both on one record is the core idea. The rest of this document is the mechanism for doing that without cheating.

What it does not claim: that the model is under version control. Model weights live with the provider, and the provider changes them on its own schedule. The record can pin three things: which model profile was used, what was sent to it, and what came back. That is enough to replay the system exactly. It is not enough to reproduce the model's choice. Section 2 takes that limit seriously.

Two git words carry over directly. A **commit** is an event on the record with a reason attached. A **branch** is a replay from some point in the record with one thing changed. Two git words do not carry over. There is no **merge**, because the record is append-only and the agent is one line of history rather than many. There is no **rebase**, because rewriting the past is the one thing this design refuses to do.

Figure 1.
The fix for both: put the agent in **version control**. Its definition and its whole history, with every change pointing at the reason.

ASK YOUR AGENT

```
I am reading a primer that claims this runtime puts "the agent in version
control": the agent's definition and its whole history live in one append-
only record, and acting is the same thing as committing. In this repository,
show me where an agent's definition is declared, where the record is written,
and whether there is any code path that performs an external effect without
first appending to the record. Cite file paths.
```

2. Two computers

"The "I'm sure" computer and the "maybe" computer."

An agent system contains two kinds of computation. They should never be confused.

The first kind is ordinary software. Given the same inputs, it produces the same outputs. You can test it, rerun it, and check it. Call it the “I’m sure” computer.

The second kind is the model. Given the same input, it may choose a different answer. Its provider can change it underneath you. Its output depends on sampling, on the state of the service, and on a model version you do not control. Call it the “maybe” computer.

Most agent frameworks mix the two together. The model’s answer lands in a variable. The variable feeds some code. The code calls a tool. The whole thing reads like one program. It is two programs, and half of it cannot be rerun.

The right response is to draw the line between the two computers on purpose, and to let that line decide what the runtime must remember and what it may recompute:

- Anything the “I’m sure” computer produced can be thrown away and rebuilt from its inputs. That covers indexes, dashboards, search tables, and the context window assembled for a model call.
- Anything the “maybe” computer produced, and anything the outside world did, cannot be rebuilt. It must be written down the moment it happens. That covers model outputs, the payment API’s reply, a user’s message, and a process exiting.

Every harness already follows half of this rule. Everyone saves what they sent to the model, because everyone has learned you cannot recompute a model’s answer. This rule extends that habit to the whole system. That extension is where the payoff is. You can only rerun yesterday without paying for model calls again if the model’s outputs are part of the record the system runs from. You can only explain what happened between model calls (routing on a classification, searching an index) if the computed state the code read is also on the record. A prompt log explains the model. It cannot explain the system, and you cannot restart a system from its prompt log.

There is an older name for this arrangement. In the theory of interactive proofs, Merlin is a prover with unlimited power who cannot be trusted. Arthur is a verifier with limited power who can check Merlin’s claims but cannot produce them [1, 3]. The model is Merlin. The runtime is Arthur. The record is the transcript Arthur keeps. Anything the system later says about what the agent did is read from that transcript. Nobody asks Merlin again.

This division also keeps claims honest. A runtime built this way can prove things about ordering, authority, mediation, and whether the record is complete. It cannot prove that what the model said was good or true, and it should not pretend to.

Discussion

One common objection: a model at temperature zero is deterministic, so the line is in the wrong place. It is not. Sampling is only one source of drift. Provider-side revisions, batching, hardware, and the serving stack all move the output over time, and you control none of them. “Same input, same output” is a promise the runtime can make about code it ships. It is not a promise anyone can make about a hosted model, so the design does not rely on it.

A second objection is cost. Writing down every model output, every external reply, and every user message sounds like a lot of storage. It is less than it sounds. The things that cannot be recomputed are small: text in, text out, an API reply. The things that are large (indexes, embeddings, assembled context windows) are exactly the things the rule lets you delete and rebuild. The line between the two computers is also the line between what you must keep and what you may throw away.

The most important consequence is about where judgment sits. The “maybe” computer may propose anything. The “I’m sure” computer is the only one allowed to make something happen, and it acts only on a written proposal it can show you later. Every section after this one is a variation on that pattern. The model proposes. The runtime decides. The decision leaves a receipt.

LINEAGE optional reading

Interactive proofs, and the thing none of the ancestors had. The prover-and-verifier split comes from complexity theory. Babai’s Arthur and Merlin games (1985) [1] describe a weak, honest verifier accepting claims from an unboundedly clever, untrusted prover, as long as each claim comes with a certificate that is cheap to check. Shamir’s 1992 result, IP = PSPACE [2], says that such a verifier, given interaction, can check far more than it could ever compute on its own. Read as a design claim: a small deterministic runtime can govern an arbitrarily strong model, provided every consequential step arrives as something checkable. A receipt is that certificate. Proof-carrying code [4] makes the same move for untrusted programs.

The analogy has a limit. A theoretical prover finds the accepting path by definition. A model only samples, and sometimes proves nothing. That is why budgets exist (there is no fixed point to wait for), and why the construction rests on the everyday gap between generating and checking, not on any unproven conjecture. Every older tool this design borrows from (type systems, effect handlers, partial evaluators) assumes its functions are deterministic. Verlet adds one typed slot for a function that is not: the model-backed step. The rules in sections 3 through 5 are the conditions under which that slot can sit inside otherwise lawful machinery.

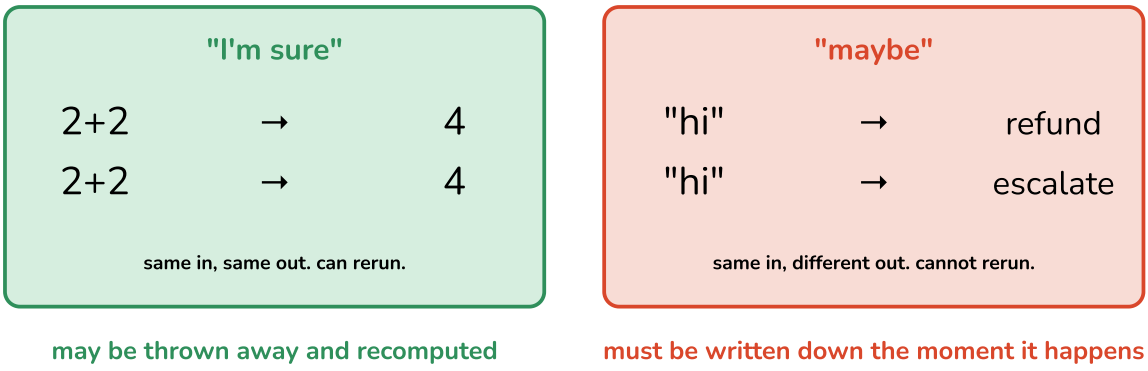


Figure 2.

Indexes and dashboards are green. Model answers, payment replies, and user messages are red. **Red goes in the notebook. Green is a cache.**

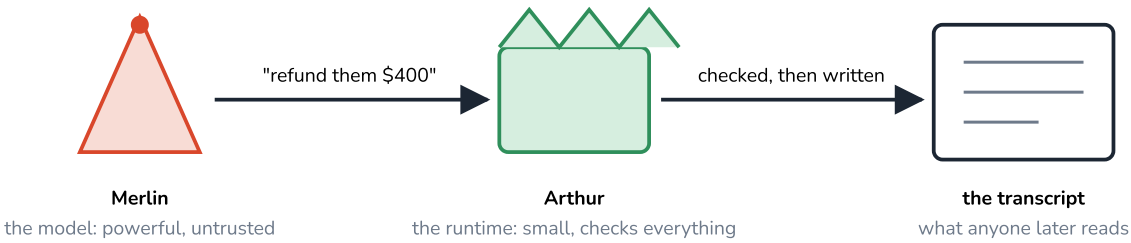


Figure 3.

Nobody asks the wizard again what he said. They read the transcript Arthur kept. That transcript is what **attestation** means.

ASK YOUR AGENT

```
This runtime distinguishes outputs that can be recomputed (indexes, views,
assembled context) from outputs that cannot (model completions, external API
replies, user messages). Find the type or enum that encodes that distinction,
show me how a model completion is persisted as an event, and show me one
thing that is treated as a disposable view. Explain the rule the code uses to
decide which is which.
```

3. The record, shown

“Write it down.”

The rule is one everyone was taught and no agent framework kept. Everything the system sees and everything it concludes goes into one append-only record, and nothing is ever erased. Here is the exact form that takes.

Streams and events

A **stream** is an append-only, ordered sequence of events that belongs to one scope. The support agent’s conversation with one customer is a stream. The control channel that configures the agent is another stream. The audit scope of the tenant that owns the agent is a third. Streams are never edited. Everything else in the system (every index, every dashboard, every context window) is computed from streams, and nothing else is kept permanently. Order within a stream is the sequence number, assigned when the event is appended. There is no clock to disagree with it. This is the old lesson about ordering in distributed systems [24].

There is no session object, no scratchpad, and no mutable “agent state” sitting beside the record. The agent does not have state. It has a history, and state is one way of reading that history.

An **event** is one permanent fact on a stream. Events have one of two origins.

A **witnessed** event records something the world or the runtime did. A customer message arrived. The payment API returned. A process exited. A model call completed. The runtime saw it happen, and that is what gives the event its authority. Witnessed events are the only way anything enters the system from outside.

A **produced** event records something a component computed. This ticket was classified as a refund request. This conversation was summarized as follows. Every produced event carries **provenance**: the events it was computed from, and the function and version that computed it. A produced event that cannot name its inputs is not written.

One rule connects the two kinds. A later event may replace an interpretation. No event replaces history. When the classifier changes its judgment, the change is a new event that points at the old one. The original judgment, and the fact that the system acted on it for an hour, stay on the record forever.

Receipts

A **receipt** is a recorded explanation of something the runtime resolved. Which human-readable tool name resolved to which immutable operation. Which attachment made a tool visible to this thread. Which policy decision authorized this action. Which sources went into the context the model saw. Receipts are written under the code and configuration in force at the time. They are never recomputed later under newer code, because a recomputed receipt would answer a different question.

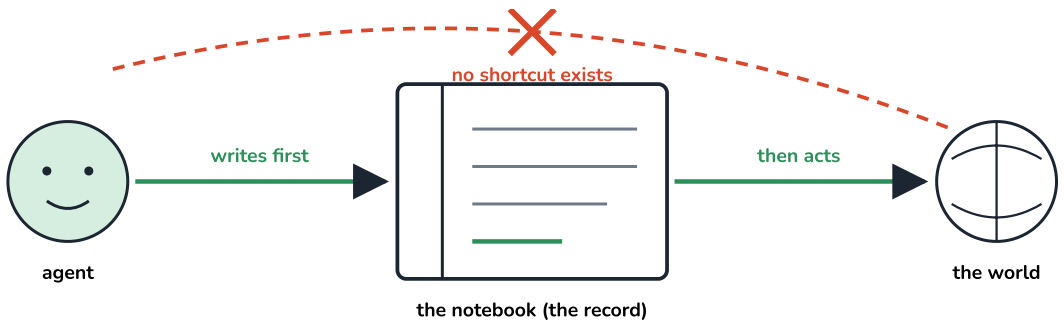


Figure 4.
The notebook is the only door. Acting **is** writing it down, so the record cannot have holes.

LINEAGE optional reading

Algebraic effects and handlers. In the effect-systems tradition [12, 13], a function does not perform its side effects. It declares them, and a separate handler chosen by the caller decides how each effect is carried out. A tool in Verlet is built the same way. The operation declares what it needs (an HTTP origin, a secret by name, a durable write) and cannot reach any of it directly. It asks through an opaque handle. The runtime is the handler, and it chooses at call time how the effect is carried out. This is why the same operation runs unchanged on a laptop and on a managed host: same program, different handler stack. Nested calls narrow rather than widen. A child operation can use at most what its parent had and what the child itself declared. The departure from the textbook is the receipt. Academic effect systems handle effects. This runtime handles them and writes down that it did, with the evidence. An effect with no receipt did not lawfully happen.

What it looks like

Here is the refund as the record has it. One customer thread, one turn, event kinds named as the current runtime names them. Payloads are trimmed.

```
seq  kind                         payload (abridged)
41   turn.submitted               principal=operator:support-queue
                                  message="Order 8812 arrived broken, refund please"
42   context.compile.completed    sources=[instructions@v3, thread:1-41,
                                  memory:customer-8812] tokens=2,114
43   session.entry.appended       role=assistant profile=support-default
                                  output=tool_call(payments.refund,
                                  {order:8812, amount:400.00})
44   tool.call.requested          op=payments.refund@sha256:9f3c…
                                  binding=b-201 args_fp=7e11…
45   tool.call.decision           outcome=hold rule=refund-over-250
                                  policy=finance-guard@v7
46   approval.requested           to=principal:finance-oncall
                                  facts={op, args_fp, principal, rule}
        … 14 minutes …
47   approval.resolved            by=principal:alice outcome=allow
48   tool.call.completed          witnessed: payment_api 200
                                  txn=pay_01J…  effect_class=at_most_once
49   session.entry.appended       role=assistant
                                  output="Refunded $400 to your card…"
50   turn.completed               reason=quiescent
```

Read it top to bottom, and every question from section 1 has an answer in the data. What happened: line 48. Why: line 43, with the context that produced it at line 42. Who allowed it: line 47, by name, against the rule at line 45. What the agent could even see: the sources at line 42 and the binding at line 44. None of this was reconstructed afterward from three databases and a log file. The runtime wrote each line because it needed that line to do the next thing.

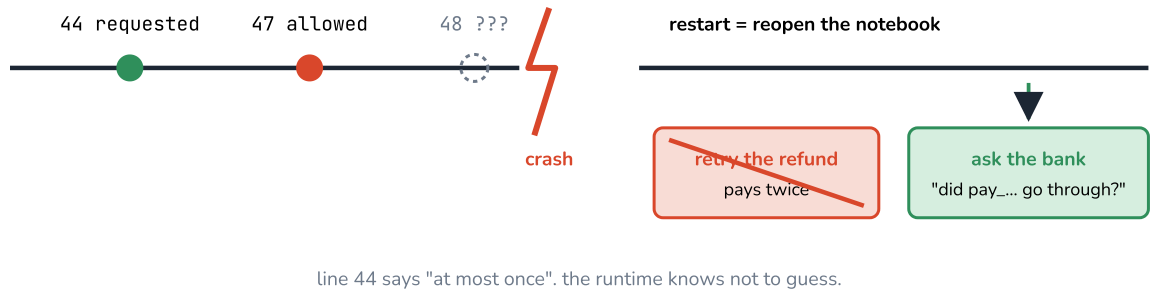
Figure 5.
What happened? Line 48. Why? Line 43. Who allowed it? Line 47, by name. Nobody had to reconstruct anything.

Why recovery is free

The true state of the system is computed by reading the events in order. So a restart is a reading problem. Reopen the streams. Rebuild the disposable views. Restore the current attachments. Continue from the next allowed step.

Take the scene everyone dreads. The process dies after the payment call at line 48 goes out and before its result is recorded. On restart, the runtime finds a `tool.call.requested` at line 44, an allow at line 47, and no completion. The operation's declared effect class is at-most-once. So the runtime does not retry. It asks the payment API for the transaction by the request's fingerprint, records what it finds as a witnessed event, and then either continues or stops with an explicit interruption. It does not guess, and it does not refund twice.

(In a real production system you should also make the refund API itself idempotent, keyed on the request fingerprint. Then the operation can be declared idempotent, and recovery becomes a safe retry instead of a lookup. The runtime's job is to know which of the two cases it is in, and to record which one it did.)



line 44 says "at most once". the runtime knows not to guess.

Figure 6.
Recovery means **reading the record**. No model call, no double refund. (And make the refund API idempotent too.)

Compare a harness whose state lives in memory and whose log is a side channel. After the crash it has a transcript and nothing else. Its honest options are to replay the whole conversation through the model again (paying again, and possibly getting a different decision) or to ask a human what happened.

Why replay is exact

Every model output and every external reply is on the record. So the deterministic parts of yesterday can be rerun against it exactly, with no model calls. You can branch at line 45, swap in `finance-guard@v8`, and see whether it would have held the refund. The cost is reading the stream. This is what makes a change to the agent testable before it ships, which is the point of putting the agent in version control.

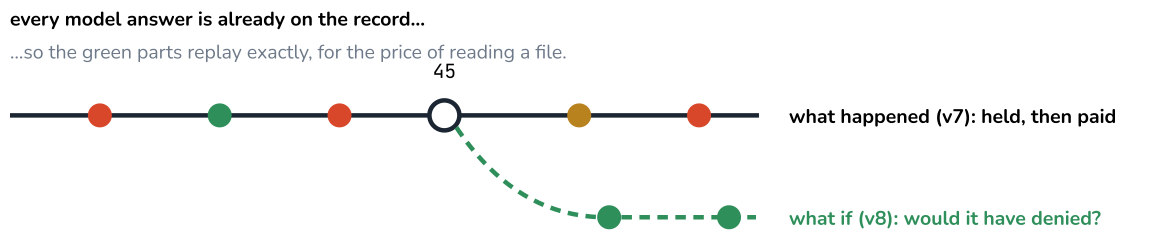


Figure 7.
This is how a change to the agent gets **tested before it ships**. Version control needs branches, and now the agent has them.

Why observation is not enough

A monitoring product watches an agent from the side and infers what it did. That is useful. It is also a different thing. Important steps may be missing, sampled, or disconnected from the state that governed them. The result is a best guess about the system.

A runtime where acting is committing produces the evidence while doing the work, because it could not have done the work otherwise. Monitoring products stay useful as views over that record. They simply start from a stronger source: the facts that execution itself required. Section 6 comes back to this, because it is the line that matters most when someone has to certify an agent.

Discussion

Three questions people ask at this point.

Can the record be wrong? It can be incomplete, if a process dies between a side effect and the event that witnesses it. That is the crash scene above. The answer is idempotent effects plus replay, not a promise that it never happens. What the record cannot be is silently edited. Events are appended under the authority of whoever produced them, and the runtime's own statements are marked as such. A wrong entry is corrected by a later entry that says so. Nothing is overwritten.

Is this just logging with extra steps? The pattern is event sourcing [23]. The difference from logging is which way the dependency points. A log is written after the fact, from state that lives somewhere else. If the log is lost, the system keeps running. Here the record is the state. A thread's tool list, its budget position, its held actions: all of it is computed by reading the record forward. If the record were lost, there would be nothing to run. That is what makes replay exact and recovery free, and it is a property that a log, however thorough, cannot offer.

What is a receipt not? It is not proof that the decision was right. A receipt for a policy decision says which policy, which inputs, which outcome, and under which code. Whether the policy was a good one is a question for the people who wrote it. The receipt is what lets them ask that question with the facts in hand.

ASK YOUR AGENT

```
Walk me through one turn of a thread in this runtime as a sequence of events:
the kinds turn.submitted, context.compile.completed, tool.call.requested,
tool.call.decision, approval.requested, session.entry.appended,
approval.resolved, tool.call.completed, turn.completed. For each, where is it
appended, what is in the payload, and which ones carry provenance pointing at
earlier events? Then show me what resume does after a crash: where does it
reopen the stream, and how does it decide whether an in-flight tool call is
retried, looked up, or failed? Find the effect class concept (at-most-once vs
idempotent) in the code.
```

4. The propagator, or where the name comes from

"An agent is a loop wired into a record."

Section 3 showed the record. This section describes what runs around it. The answer is smaller than most frameworks make it.

Everything that acts on a thread is a **coupling**. A coupling is a function with three declared parts: what it reads (a selection of streams), where its output goes, and what triggers it. A coupling has no power beyond those three declarations. It reads what it named, writes where it named, and runs when its trigger fires. Every piece of machinery in the runtime, including the agent loop, is a coupling.

Couplings come in exactly three kinds. The kind is decided by one question: where does the output go?

- If the output goes back into the stream the coupling read, the coupling **advances** the system. This kind is called a **propagator**. The agent loop is the main one. It reads the thread's record, calls the model, and appends the model's answer to the same record. It is triggered by a turn being submitted. Thread in, model, thread out.
- If the output goes somewhere else as a derived result, the coupling **derives**. This kind is called a **projection**. Examples: a summary, an embedding, a classification, an index feed. Projections are allowed to lose information. That is their job. What they may not do is hide what they read, so every projection's output names its inputs.
- If the output goes into a control stream, meaning something that changes how future turns run, the coupling **steers**. This kind is called a **controller**. Examples: switching to a more careful model when a conversation escalates, gating a tool, deciding when to

compact context, routing an inbound message to a thread. Products call these hooks. They are feedback loops.

Advance, derive, steer. These three cover every case, because an output can only return to its source, land elsewhere as an interpretation, or feed forward into control. Try it on your own stack. The support agent breaks down with nothing left over. The loop is the propagator. The ticket classifier, the summarizer, and the embedding feed are projections. The escalation rule and the compaction trigger are controllers.

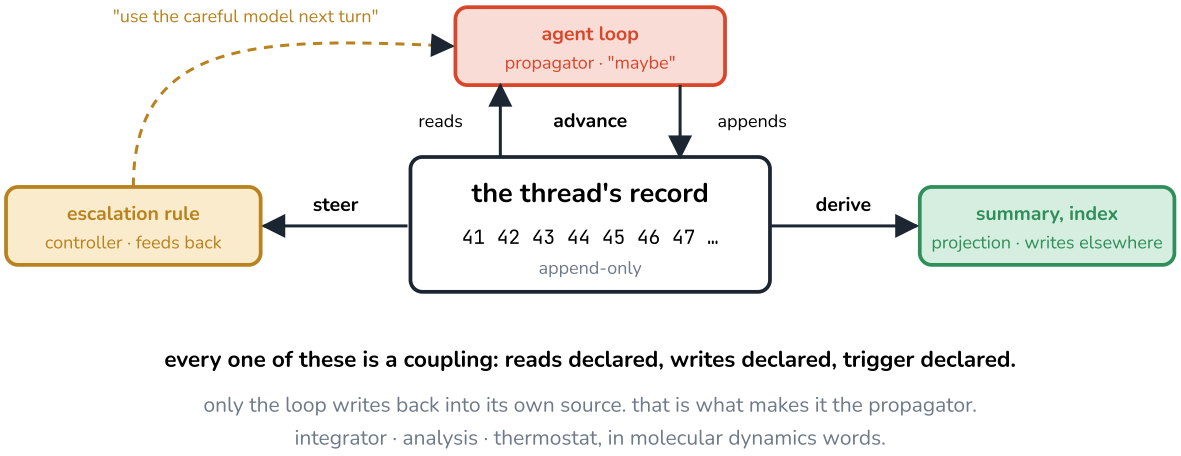


Figure 8.
One record, three kinds of function around it. The loop **advances** it. Projections **derive** from it. Controllers steer what the loop does next. The agent is the loop and nothing more.

Why this is the whole runtime

Having only one kind of thing means nothing about the agent is special. An agent is a propagator wired into streams. A sub-agent is another propagator on another thread. A workflow is a propagator whose next step is chosen by a script instead of a model. A “memory system” is a projection that writes plus a context source that reads. The runtime does not ship an agent abstraction, a workflow abstraction, and a memory abstraction that happen to work together. It ships streams and couplings. The product words are names for wiring patterns.

This also says where the two computers from section 2 live. A coupling is either deterministic or **chaotic**. Chaotic means its output cannot be reproduced from its inputs. Model-backed couplings are chaotic by definition. The rule from section 2 becomes a rule about couplings. A chaotic coupling’s output must land on a stream as an event, because nothing deterministic could regenerate it. A deterministic coupling’s output may be a view

instead, which can be cached and deleted. The agent loop is chaotic. That is why every one of its outputs is on the record, and why the checkpoint in the next section sits directly on its output.

One more property follows. Nothing in the system can start itself. Couplings fire on events. Derived events trace, through provenance, to the events that produced them. That chain only ends at a witnessed event, which came from outside: a user's message, a timer the operator set, an API reply. "The agent decided on its own to start doing something" is not a scenario the runtime can represent. Every chain of activity begins with someone.

Discussion

The word propagator is borrowed from molecular dynamics, and that borrowing is where the project's name comes from. In a molecular dynamics simulation, the **integrator** advances every particle one step from the current state. The Verlet integrator (Loup Verlet, 1967 [17]; the velocity form in common use is from 1982 [18]) is the standard one. It has a property that matters here. It is time-reversible, and it conserves what it should conserve over long runs, because it is built around the trajectory rather than around a single force evaluation. **Analysis functions** are computed over the trajectory without touching it. **Thermostats** and biases read the trajectory's history and feed back into the dynamics to steer them. Advance, derive, steer, one-to-one [20]. (The word is used in the integrator sense, not in the sense of the propagator networks of Sussman and Radul [26], which are a different idea.)

The creator's training is in computational chemistry and physics. Verlet began with the observation that the abstractions that make a molecular simulation trustworthy are the same ones an agent runtime needs and usually lacks: one trajectory as the truth, an integrator that only ever advances it, analysis that never writes into it, and feedback that is itself a function of the history. The mapping is a memory aid, not a proof. The definitions above stand on their own. But the name is a reminder of the discipline. The thing that advances the system is one small, well-understood function, and everything else either reads or steers.

LINEAGE optional reading

Metadynamics, or why agent memory is a thermostat. Molecular simulations have a standard problem. The system gets stuck in one valley of its energy landscape and never visits the rest. Metadynamics (Laio and Parrinello, 2002 [19]) fixes this with a bias that is a function of the trajectory's own history. Every so often it drops a small hill where the system currently is, so that places already visited become less attractive and the system is pushed toward what it has not yet seen.

Two things about that construction carry over. First, the bias reads the history and writes into the dynamics. It is a controller in the sense above, and agent memory has exactly that shape: a projection over past turns whose output changes what the loop does next. Second, and this is the part most memory features miss, the bias is a deterministic function of the trajectory. Given the record, you can recompute what the memory was at any step. That is what makes its influence explainable after the fact. The lexicon's line for this is short: agent memory is metadynamics, a history-dependent bias shaping future evolution. The departure is the usual one. A thermostat does not need receipts, because a simulation has no adversary and no auditor. Here the feedback leaves events with provenance, so "why did it pick the careful model on turn 12" is a query rather than a guess.

ASK YOUR AGENT

```
This runtime classifies every function that acts on a thread as a coupling
with declared reads, writes, and trigger, and sorts couplings into
propagator, projection, or controller by where the output goes. Find the
coupling definition and the three roles in the code. Show me the agent loop
as the privileged propagator (thread in, model, thread out), one projection,
and one controller, and show me where a chaotic coupling is forced to write
events rather than a view.
```

5. One checkpoint

"What may happen next?"

Workflows and agents are usually sold as different products. A workflow follows a graph someone drew. An agent lets the model decide. Teams end up running both, on two stacks, with two logs, and a hand-off between them that nobody can audit.

In operation they were never different things. Every step the system takes is a continuation of some thread. That holds for a model call, a tool call, a spawned child, and the end of a turn. Put one checkpoint in front of all of them. At each step, the checkpoint asks one question: given this thread's history, which continuations are allowed next? It picks one, and it writes the decision to the record as an event like any other.

Two properties make that checkpoint more than a dispatcher. First, there is no side door. Every continuation the runtime executes passed the checkpoint, because the checkpoint is where execution is obtained. Second, the decision itself is history. The record shows more than the fact that the refund tool ran. It shows that at step 45, under `finance-guard@v7`, the allowed set was `{hold, deny}` and `hold` was chosen. A post-mortem no longer has to reconstruct what the system was choosing between.

The checkpoint's question is answered by a **policy**. A policy is a function from the thread's history to the set of allowed next steps. Policies come in two modes, and the two modes are the whole taxonomy.

In **strict** mode the policy is deterministic. Only the moves it names are allowed, and replaying the record re-derives every decision it ever made, exactly. A strict policy is what the industry calls a workflow.

In **adaptive** mode the model chooses the continuation, inside a declared envelope of tools and budgets. The choice lands on the same record through the same checkpoint. An adaptive policy is what the industry calls an agent. The everyday agent loop ("the model decides what to do next, with its tools available") is the simplest adaptive policy: allow the whole envelope at every step. It is the correct default and the least interesting setting.

Because the choice is made per step, it stops being an architecture and becomes a dial. The support agent's refund path runs strict: verify the order, check the amount, hold above a threshold. The conversation around it runs adaptive. Same thread, same record, same runtime.

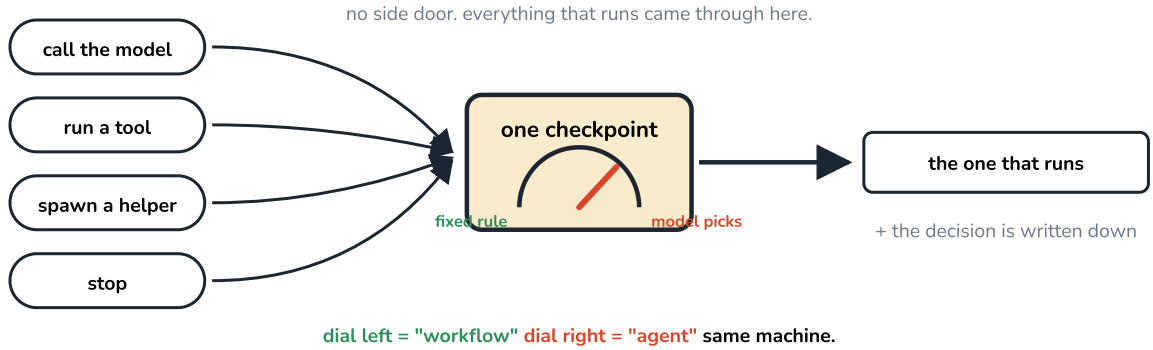


Figure 9.
A workflow and an agent are one dial, set **per step**. The refund path runs strict, the conversation runs adaptive, both on the same notebook.

Budgets instead of assumed convergence

A deterministic workflow can often prove from its graph that it terminates. A model-backed loop cannot. So the runtime enforces explicit limits on turns, depth, time, cost, and triggered work. A turn ends when the policy reaches quiescence (nothing left to do) or a budget stops it. Budgets are runtime mechanisms, not instructions in a prompt, and they leave evidence. The record says whether work completed, was stopped by policy, ran out of budget, or was canceled by a named person.

Discussion

Treating strict and adaptive as one dial, set per step, changes how a team builds. Nobody has to decide up front whether a task is “an agent problem” or “a workflow problem.” A support flow can be strict for the refund step (three checks, in order, no improvisation) and adaptive for the conversation around it. The mix is a property of the policy bound to the thread, and the record shows which setting governed each step.

Budgets are easy to dismiss as configuration. They are the only honest answer to a loop with no fixed point. A deterministic workflow stops because its graph runs out. A model can always propose one more step, and “it will probably converge” is not a guarantee anyone should ship. A budget turns an open question into a bounded one. And the record’s statement of why a turn ended (done, held, out of budget, canceled by whom) is the first thing an operator reads when something went wrong.

Improvisation becomes procedure

A policy is data, so an agent can write one. Suppose the support agent has improvised the same three-step check on fifty broken-order tickets. That sequence can be published as a strict policy, with provenance pointing at the fifty episodes, and bound for future runs. The model stays available for the novel case. The repeated case gets cheaper, faster, and easier to inspect. This is the path from improvisation to procedure, and the record supplies the reasons behind the diff.

LINEAGE optional reading

Partial evaluation and the Futamura projections. Partial evaluation asks of every part of a program: when does this become known? Anything known early can be specialized away, leaving a smaller residual program for what is still open. Futamura's 1971 observation [5] was that applying this to an interpreter and a program yields a compiled program, and applying it again yields a compiler. Improvisation becoming procedure is the same move one level up. The running agent is an interpreter. What it interprets is the task, in natural language. Repeated interpretive work (the same three checks on fifty tickets) is the part that has become known, and condensing it into a strict policy is specialization with a receipt.

The deterministic machinery below (manifest in, bound harness out) is the first Futamura projection in the textbook sense. The specialization above it is chaotic, because a model proposed it. That is exactly why it lands as a proposal on the record and passes the checkpoint instead of being assumed. There is even a gauge. Jones optimality [6, 7] asks whether specialization removed the entire interpretive layer. Here, the share of a thread's steps that still went to the model, per recurring task, is that overhead, and it can be read straight off the record. Inference systems already do this by hand: prefix caching is the model specialized on a fixed context, and fine-tuning is weights specialized on history. The rule this ancestry leaves behind is that specialization must be auditable. A residual program with no receipt for how it was produced cannot be trusted.

ASK YOUR AGENT

```
The primer says every continuation (model call, tool call, child thread, end
of turn) passes one admission checkpoint, and a policy answers "what may
happen next" in either a deterministic mode or a model-selected mode. Find
that checkpoint. Is there any way for a continuation to execute without
passing through it? Show me where budgets (turns, depth, time, cost) are
enforced and what event is written when a budget stops a turn.
```

6. On whose authority?

“A tool that is not attached does not exist.”

Here is the failure that governance products exist to prevent. The model proposes a refund of $40,000. Nothing checked whether it was allowed to. The money moves. Afterward, the team discovers that the refund tool was reachable from a prompt nobody had reviewed, through a path nobody had drawn.

The dangerous part was never that the model proposed it. Models propose things constantly. The dangerous part is an unguarded route from the proposal to the world. Two laws close that route. Both are built into the runtime, not configured on top of it.

Attachment. An operation that has not been attached to a thread has no tool surface on that thread. It is not “denied”. It does not exist. It is not in the model’s tool list and cannot be named. Attaching is an event on the record (section 7 covers it). So the question “what could this agent have done?” is answered by reading the thread’s attachments, not by reading prompts and hoping every branch was found.

Mediation. Every external effect passes through one runtime-controlled surface. That surface is where a configured policy can allow, deny, or hold the action. It is also where the decision is recorded and tied to the action it governed. A denial is visible to the agent, so it can choose another path. An approval authorizes exactly one fingerprinted action, not a class of actions.

Behind both laws sits a **principal**: the human or service identity on whose authority the work entered. Line 41 of the excerpt names one. The agent is not itself a principal. It acts within authority that arrived from outside, and every effect traces back to the name that asked for it.

Why rules alone cannot do this

Rules-first governance keeps failing at agents, and the reason is older than agents. Ashby’s law of requisite variety [8, 9] says a regulator can only control a system whose variety it can match. A policy written in advance cannot match the variety of what a model will propose. Every team that has tried to list the allowed actions in advance has watched the list go stale within a week.

The regulator that can match model variety is not a longer list. It is the pair above: a complete record, and one choke point that every effect must pass. At that choke point, a decision of any sophistication can be made, by a rule, a person, or another model, at the moment the specific action is known. The runtime supplies the choke point, the hold, the resume, and the record. The judgment is pluggable.

Discussion

Attachment as the unit of authority takes some getting used to. Most systems make authority a property of the agent: “this agent is allowed to refund.” Here it is a property of the thread’s history: at sequence 12, a refund tool was attached, with this configuration, on this person’s authority, and it has not been detached since. The difference shows up the first time someone asks “since when could it do that, and who said so?” A role-based system has to reconstruct who changed the role and when. An attachment-based system answers with one query.

Mediation is the other half. It is tempting to picture the checkpoint as a firewall that inspects traffic. It is closer to a notary. The action does not happen and then get checked. It is proposed, held, decided, and only then performed, and the decision is written before the effect. This ordering is what turns “every consequential action was decided” from a claim about coverage into a claim about construction.

LINEAGE optional reading

Cybernetics. The regulation vocabulary here (controllers, feedback, budget-enforced quiescence) descends from cybernetics [10]. Ashby’s law of requisite variety is the one theorem doing real work: a regulator must have at least as much variety as the thing it regulates. A fixed checklist dies when the model’s behavior outgrows it. A checkpoint that sees the specific proposed action, with the whole record available, grows with the model it governs.

The interesting part is what cybernetics lacks. It has no notion of truth (an authoritative, append-only account of what happened), none of attestation (who witnessed a fact versus who produced it), and none of permission (what a part *may* do, as opposed to what it *can* do). Every Verlet deployment is a cybernetic system. Almost no cybernetic system is a lawful Verlet one, because those three commitments are what is left after the dynamics are accounted for. Even second-order cybernetics [11], which put the observer inside the system, left observation as a signal. Here the act of observing is itself a durable event with provenance. The regulator must leave receipts. It is Arthur, not merely a thermostat.

Governance should keep a path to yes

A control that can only block pushes people toward bypasses. Useful governance offers a way forward. Say why the action was denied and which rule applied. Let a person approve with a bounded set of facts. Let the agent revise the action. Detach a failing capability. Record an authorized exception. Resume held work after a decision. Lines 46 and 47 of the

excerpt show that path in miniature: fourteen minutes of a human deciding, with the thread parked and nothing lost.

Bounded disclosure

The person approving a refund rarely needs the whole transcript. A decision request carries a typed fact sheet: the action, its argument fingerprint, the principal, the binding, the rule, and a bounded slice of history. Any explanation the agent wrote is marked as untrusted content. Less private context leaves the thread, and text the model generated cannot quietly present itself as authority.

By construction and by observation

This is where the two-computers cut pays off for anyone who has to certify an agent. There are two ways to know what an agent did.

By observation. A sensor sits beside or in front of the agent, captures what it can see, and reconstructs a chain of custody. This is what most governance products do today. It works on any agent, which is its strength. It is an inference, which is its limit. It cannot know what the agent could have done, only what it saw it do.

By construction. The runtime is the only way to act, so the record is not a reconstruction of the work. It is the work's precondition. The evidence exists because the action could not have happened without it. Attachment answers "what could it do." Mediation answers "who allowed it." The record answers "what did it do." All three come from the same source.

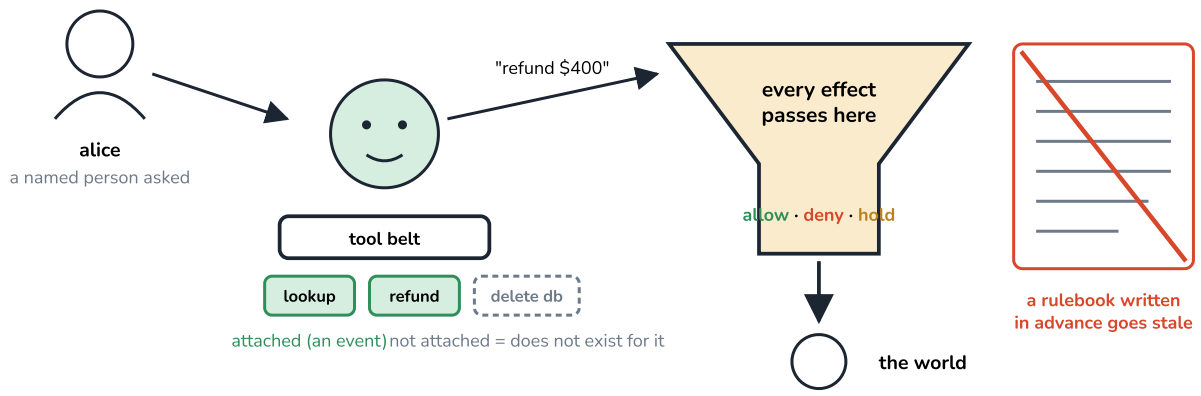


Figure 10.
Two laws, built in: a tool that is not **attached** does not exist, and every effect passes through one **door**. The judgment at the door can be swapped. The door cannot.

The two compose. An observation product can consume a by-construction record as its strongest input, and map it to the controls and frameworks its customers need. The integration is two seams. Policy decisions come in: the runtime sends a structured request and records the answer. Runtime evidence goes out: events and receipts are exported with stable identities and links back to the source record. The runtime stays the place where a decision becomes an enforced and recorded action.

Honest status: attachment and mediation are built into the shipped runtime. The pluggable policy router, with hold and resume behind an external service, is designed and not yet shipped. Section 10 keeps the full ledger.

Figure 11.
Watching from outside gives a good guess. The single door gives proof. The two stack: an auditor reads the proof through the monitoring dashboard.

ASK YOUR AGENT

```
Two claims to check. First: an operation that is not attached to a thread has
no tool surface on that thread, meaning it does not appear in the model's
tool list at all. Show me where the tool list for a model call is computed
from attachments. Second: every external effect passes through one runtime-
mediated surface where a policy can allow, deny, or hold it. Show me that
surface, and show me how a hold is parked and later resumed. Also show me
where the principal (the identity on whose authority work entered) is
recorded on a turn. Tell me honestly which parts of the pluggable policy
router are implemented and which are not.
```

7. The harness is data

"Don't put it down, put it back."

Section 1 promised the agent in version control. Sections 3 through 6 built the record, the loop, the checkpoint, and the authority model. This section is where the promise is kept.

Everything that makes up the agent's harness is declared data: which tools it has, which policy governs it, which model profile it runs, which sources feed its context. Every change to that data is an event on the thread's record.

A tool becomes available through an **attach** event and unavailable through a **detach** event (`binding.attached` and `binding.detached` in the record). The set of tools the agent has at any moment is not stored anywhere. It is computed by reading those events up to now. Configuration for a tool (which secret it may use, which network it may reach) rides on the attach event. So "what could it reach, and since when" is one query over the record.

Changes take effect at a turn boundary, going forward. A turn already in progress keeps the snapshot it started under. A registry entry changing tomorrow does not alter the contract a running turn started with today, because execution resolved the human name to an immutable, content-addressed operation and recorded that resolution. Rollback is a new recorded change, not a rewrite of the past.

Self-extension through the same gate

Now the 3am scene. The support agent, working a ticket, discovers it needs a lookup it does not have. It writes the operation (a small script, published as an immutable package) and requests attachment.

That request is an action like any other. It reaches the checkpoint. A policy decides whether the change widens the agent's authority, merely adds structure inside authority it already has, or needs a human. The attach event records who requested the change, which decision authorized it, and which immutable operation became available. At the next turn boundary, the agent's tool list reflects it.

Nothing new was needed for this. The same record, the same checkpoint, the same mediation, and the same receipts that govern a refund also govern the agent changing itself. That is the entire mechanism. It is why the bricking story from section 1 stops. The change is a commit with a reason. The episodes that motivated it are on the record next to it. Detaching it is one more event.

Version control, finished

Put the pieces together and the git analogy is complete. The agent's definition is data under history. The agent's experience is the same history. Each change points at the experience that motivated it. Trying a change is a branch at a sequence number with a different policy bound. A bad change is a detach. Nothing is ever off the record, because the record is how anything happens.

Discussion

"The harness is data" can sound like "the harness is a config file," which every framework already has. Three properties separate the two.

First, the harness is never edited in place. It is computed from attach and detach events, so every version of it that ever existed is still readable, and each change sits next to the reason for it.

Second, the harness is resolved as well as declared. A manifest proposes tools by human-readable name. What the thread actually runs against is the set of immutable, content-addressed operations those names resolved to at bind time, with the resolution receipted. Exporting the harness means reading that resolved set. That is why two machines can compare harnesses and find the exact line that differs.

Third, the agent can change the harness only through the same gate that governs everything else. This is the property that makes self-extension safe enough to allow at all. There is no privileged "reconfigure" path that bypasses the checkpoint. So there is no way for the agent to quietly become something other than what the record says it is.

LINEAGE optional reading

The program is data; declarations select, they do not compute. Two older schools meet here. The first is the initial-encoding school in functional programming [14, 15]: represent the computation as a value, keep the interpreter swappable, and enforce the laws in the constructors that build the value. The manifest is that value and the kernel is its interpreter. The constructors are the compile and bind steps. Names resolve to immutable hashes, unknown references are rejected before anything runs, and every resolution gets a receipt. Hot-swap works because reconfiguration hands the interpreter a new term. It never patches the interpreter. The rule this leaves behind: if a behavior cannot be read off the declaration plus the receipts, it is hidden interpreter state, and a bug.

The second school is template metaprogramming. Declarations are templates and bind is instantiation, done fully before any run. One operation definition instantiates as a CLI command, an HTTP route, a model-visible tool, and an MCP export, each faithful to the same definition. The exported harness is the expansion, the post-instantiation artifact that shows what the declarations became. C++ templates became an accidental programming language with no laws, no budget, and unreadable expansion errors. Verlet caps the declaration layer on purpose. Declarations select and arrange among registered pieces. Anything that computes is an operation, where attachments, provenance, and budgets apply.

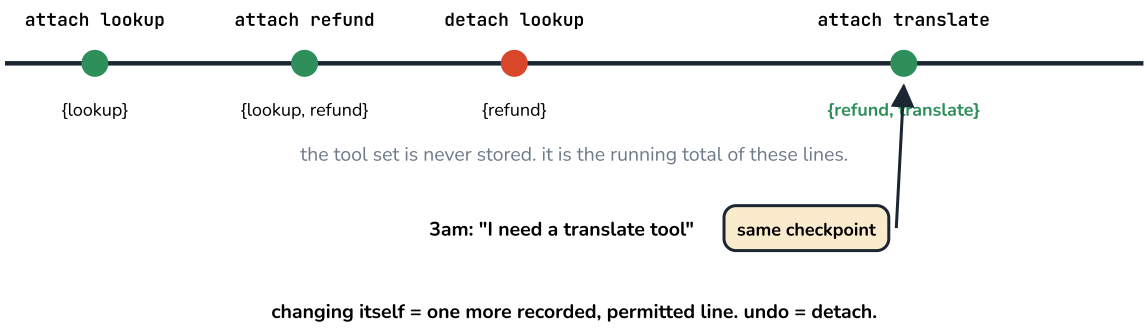


Figure 12.
The agent's harness is **data on the record**. When the agent changes itself, the change goes through the same door as a refund.

ASK YOUR AGENT

```
Show me the attach and detach events for operations on a thread, and the code
that folds them into the current tool set. Confirm the tool set is derived
from the record rather than stored as mutable state. Then show me how a
change takes effect at a turn boundary rather than mid-turn, and how
execution resolves a human-readable operation name to a content-addressed
immutable identity and records that resolution. Finally: can an agent publish
a new operation and request that it be attached to its own thread? Trace that
path and say which policy decides.
```

8. Where it runs changes nothing about what it may do

“Mind your own business.”

Authority lives in attachments on the record. Placement is a separate decision: which machine, which sandbox, which worker. Moving a thread from a laptop to a managed host, or from one worker to another after a failure, does not change what the thread may do. Nothing about its authority was ever stored in the machine.

Two consequences follow. First, many customers’ agents can share one runtime host. Each sits inside its own fence, and the fence is defined by attachments and principals rather than by process boundaries. Second, a sandbox and a policy decision answer different questions, and the record says which guarantee applied. A strong sandbox does not decide whether an action was authorized. A strong policy decision does not confine arbitrary code. Agent systems need both.

Discussion

The split between placement and authority is what makes the rest of this document portable. Suppose “what it may do” were a property of the machine: this container holds the payment key, so whatever runs in it can refund. Then moving the agent would mean re-deriving its authority, and a multi-tenant host would be one misconfiguration away from a cross-customer refund. Keeping authority in attachments means the machine can stay simple. It runs what it is given, against the record it is handed. The question “could this thread reach that system” is answered by the record, whichever worker happened to be executing.

The sandbox point is the one people most often collapse. A sandbox answers “what can this process physically touch.” A policy decision answers “was this action authorized, by whom, under which rule.” A system with only the first cannot explain its own behavior. A system with only the second cannot contain code it did not write. Both guarantees are worth having, and the record should say which one applied to each effect, because they fail in different ways.

The developer loop

For the person building the agent, all of this reduces to one loop. The same contracts apply locally and on a managed host.

Define. A folder holds the instructions, tools, resources, model profiles, context policy, and the secrets and networks the agent needs. It reads as a description of the runtime object. It is checked into git and can be handed to another runtime without a tutorial.

Plan. Resolve the proposed shape without running it. The plan lists which immutable operations, which secrets, which network origins, which resources, and which policy decisions will be required, and which capabilities the chosen placement cannot honor. A system that cannot explain its requested powers before execution forces the reviewer to audit code and hope.

Run. Run locally through a CLI, a terminal console, an RPC client, or an MCP or ACP surface. All of them present the same operations and the same thread history. The local run produces the same kinds of events and receipts a managed run will.

Publish. Operations are published by content identity. A mutable name may point at one, but execution resolves the name and records the resolution. Publishing a tool does not make it visible to any agent. A later attach does, within a scope, under policy.

Promote. Pick a placement and the organization’s bindings. The managed host supplies identity, tenancy, secrets, quotas, and operations. The definition stays inspectable and exportable, which is what avoiding lock-in means in practice.

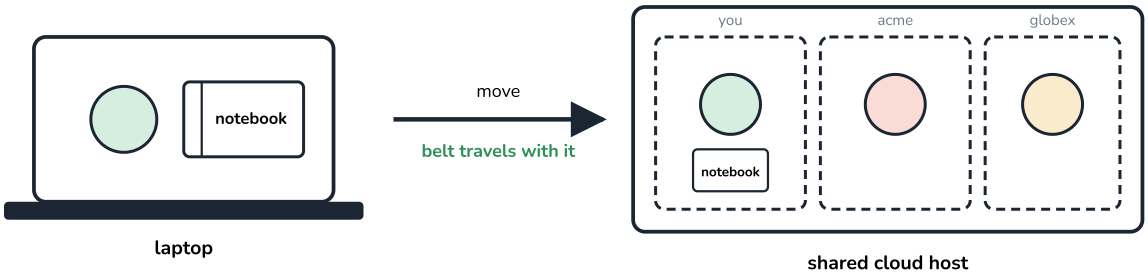


Figure 13.
What an agent **may** do lives in its notebook, never in the machine. So it can move, and many tenants can share one host, each inside its own fence.

Operate. Inspect the current attachments, the hashes that produced them, outcomes, decisions, outstanding work, failures, placement, and the provenance of any change. Revise going forward. Revoke by detaching.

ASK YOUR AGENT

```
The primer says placement (which machine or sandbox runs a thread) is a
separate decision from authority (what the thread may do), and that a thread
can move without changing what it may do. Show me the placement decision
event and where it is recorded. Show me what state travels with a thread when
it is placed remotely and what does not. Then show me how multiple tenants or
instances share one host and what keeps them apart. Also list the interfaces
(CLI, daemon, RPC, MCP, ACP) and confirm they expose the same operations and
thread history.
```

9. The field

Agent systems touch several product categories that use the same nouns while owning different things. Four responsibilities keep them apart.

Responsibility	Owns	Examples
Authoring and distribution	definitions, packages, catalogs, registries	what should exist

Responsibility	Owns	Examples
Governance and control	rollout intent, organizational policy, approvals	what should be allowed
Runtime execution	threads, attachments, mediation, recovery, placement, receipts	what ran
Evidence and assurance	search, investigation, control mapping, reporting	what it means to an auditor

The products may be bundled. The contracts stay distinct. The most important line is between desired state and executed state. A catalog says version 3 is approved for the finance team. A policy service says this action is allowed. The runtime record says whether version 3 actually started, which attachments it received, whether the decision reached the mediated action, and what followed. Both matter. They should be joinable and never collapsed into one. When they differ, the difference is the incident.

LINEAGE optional reading

Traits, typeclasses, and components. The kernel defines very few contracts: how an operation is invoked, how context is assembled, and the shape of a coupling. Product words such as “memory,” “skill,” “hook,” and “subagent” are never implemented against directly. Each is a name for a pattern over those contracts. Memory, for example, is anything with a writer that produces events and a reader that feeds context. Nothing declares itself to be a memory, and anything with that shape is one. This is the typeclass move [16] (one contract, many derived names, instances by structure rather than by inheritance) and the entity-component move from game engines, where “supports memory” is a query over components rather than a subclass. The rule it leaves behind is what keeps the four-responsibility table above stable. A new product word must lower to existing contracts. A product word that needs a new kernel primitive is a design failure until proven otherwise.

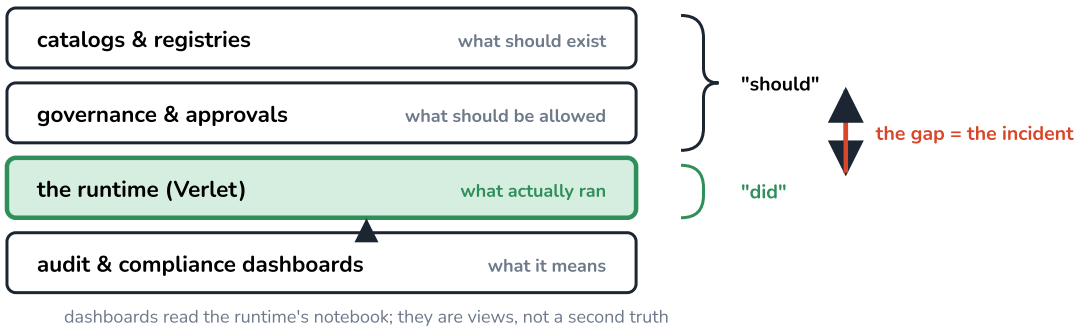


Figure 14.
Keep “should” and “did” separate but joinable. When they disagree, the difference is the incident, and it is already data.

Cordis and DeepSeek Harness

Two recent systems are moving in the same direction from a different starting point. They deserve a fair page.

Cordis [30, 31] is a TypeScript meta-framework for programs whose components arrive, leave, or change while the process is live. A component that changes shared context supplies an inverse, and the runtime tracks what it would take to withdraw it. Components declare what they require and provide, and Cordis reacts when providers appear or disappear. Its paper develops this into a calculus of components and live instances. Its results are conditional on correct inverses, confinement, and its other stated premises. External emissions cross the boundary of what an inverse can restore, and untrusted code still needs a sandbox.

DeepSeek Harness [29] applies Cordis to an agent product. Everything is a plugin: model adapters, tools, skills, sessions, storage, sandboxes, workflows, the agent loop, and the interfaces. As of 21 August 2026 it calls itself a developer preview and warns that compatibility-breaking changes will occur. Its session is event-sourced. Model-visible context is reconstructed from recorded entries, and a trajectory interface can inspect, resume, fork, and replay that history without repeating model or tool calls.

The honest contrast is about scope and custody, not about whether anyone keeps a record. Harness’s record covers the model-facing trajectory and session controls. Its composition lives in layered configuration and the live Cordis assembly. Tenant identity, principal attribution, authority over attachments, placement across a fleet, and one execution model for strict and adaptive work all need a wider runtime boundary. Cordis does have capability-style, proxy-mediated access. The narrower observation is that its formal center

is component lifecycle, not a principal and an execution record. Neither system's public materials make the tenant, principal, and placement guarantees Verlet is designed to make. That is a statement about what is claimed, not about what is impossible.

There is plenty to learn from them. Plugin boundaries have to be easy to use. The trajectory is a product surface, and fork, replay, and search make the record tangible in a way an audit API never will. The harness itself must be inspectable and changeable without reading the whole application. And an integration should target a tagged, stable surface rather than chase a pre-release contract.

Where the other products sit

Model providers supply the "maybe" computer. The runtime uses their capabilities and keeps durable identity outside any one of them. Sandboxes supply isolation, which is a placement choice, not an authority decision. Tool vendors supply operations, published immutably or surfaced through a search-and-call boundary with call-time validation. Registries distribute definitions and resolve names, while the runtime still pins the immutable identity before running. Governance and compliance products supply organizational judgment and control mapping, through the two seams in section 6. Observability and evaluation products build views over the record. A model-backed evaluation is itself a produced event with provenance. Orchestrators decide when and why agents run across a fleet, and should be clients of the runtime record rather than a second source of truth.

ASK YOUR AGENT

```
Describe the seams this runtime exposes to outside products: how an external
policy service would be called on a consequential action and what it
receives, and how events and receipts can be exported to an outside evidence
or compliance system. For each seam, say whether it is implemented, partly
implemented, or only designed, and point at the code or docs that show the
current state.
```

10. Status and receipts

Verlet is experimental and says so. Its status is kept in three separate registers, so that the architecture story does not imply production maturity.

Available in the current runtime (v0.4.0). Append-only event streams and receipts. Content-addressed operation publication. Journaled attach and detach with attachment-

carried configuration. Durable resume from the record. Agent manifest planning and publishing. Wasm operations and custom execution lanes. Virtual shell and filesystem surfaces. CLI, daemon, RPC, MCP, and ACP interfaces over shared contracts. Durable process and child-thread handles. Remote child placement through authenticated store-backed queues. Multi-instance host primitives with instance-owned authentication. (Checked against the v0.4.0 tag [28] on 21 August 2026. The vocabulary is the formalism's [27].)

In progress. The pluggable policy router with held approval and resume. Stable evidence-export profiles for external governance systems. The preset and opening-sequence authoring experience. Managed-cloud operational hardening. Richer package and distribution surfaces. Stronger placement backends.

Direction. Self-serve local-to-managed promotion. Public and private package ecosystems. Fleet control and placement mobility. Governed self-extension end to end, from proposal through evaluated deployment. Conformance batteries for the runtime and evidence guarantees. Stable bridges to external harness formats.

The criteria

The claims above can be checked. The companion paper states thirteen yes/no criteria that decide whether any system works the way this primer describes. Each can be answered from a system's records and configuration alone. The paper scores ten shipping systems plus Verlet against them. Nobody passes all thirteen, including Verlet. The paper says exactly which are met, which are partial, and which are only specified. The current self-score:

#	Criterion, in plain words	Verlet
C1	History is append-only and enough to rebuild state	partial
C2	Every event says whether it was witnessed or produced, with provenance	met
C3	Nothing deletable holds truth	partial
C4	Replay re-derives; it never re-runs the model or the world	met for resume, partial for full replay
C5	A fork is a shared prefix, nothing copied	met

#	Criterion, in plain words	Verlet
C6	The active system can be read as a wiring diagram	met
C7	Authority is attachment and mediation; not attached means absent	partial
C8	Every activation has a witnessed origin and a budgeted end	partial
C9	No side door: every continuation passed the checkpoint	met
C10	Strict or adaptive is a per-step dial on one record	specified
C11	Every tool row is a faithful surface of an immutable contract	met
C12	Every model call has a receipt for what it was shown	met
C13	The envelope exports, and placement changes nothing	partial

Six met, six partial, one specified. The runtime's test suite is organized around the same criteria, so "the record cannot have holes" points at a battery of tests rather than at a sentence.

The next proof is end to end

The next stage should be judged by one complete path rather than by isolated primitives. Author an agent locally. Publish its operations. Promote it to managed placement. Hit an external policy service on a consequential action. Record the decision and outcome. Export the evidence. Kill the runtime, and resume with the same identity and attachments. That single path tests ergonomics, runtime truth, governance integration, portability, and recovery together.

LINEAGE optional reading

Between two failure modes. Self-modifying systems in the literature sit at two extremes. The Gödel machine (Schmidhuber, 2003 [21]) requires a proof that a change is an improvement before applying it. It is formally clean and was never built. Prompt-evolution loops such as GEPA [22] and fine-tuning loops modify themselves with no durable ground truth, no receipts, and no reproducible evaluation. They are practical and unaccountable, because they cannot climb a landscape they cannot stand still on. The design here is the middle. Changes are proposals on the record, accepted through a checkpoint, with the episodes that motivated them attached, and a detach always available. There is no proof of improvement, but there is a complete account of what was tried and why. The status table above should be read with that framing. What is listed as partial is mostly the acceptance machinery for that middle path, not the record it rests on.

ASK YOUR AGENT

```
List the thirteen criteria this runtime says it measures itself against (look
for the criteria in the docs or the test suite organization), and for each
one point me at the test or battery that checks it. Then tell me which of the
capabilities listed as “available” in the primer you can confirm from a
current checkout and which you cannot.
```

11. Ten questions for any agent system

Use these without adopting Verlet’s vocabulary.

1. What is the unit a developer authors, and can the effective agent be inspected without reading application code?
2. Which facts are authoritative after a crash, and can the system restart without regenerating model output?
3. How is an ambiguous external effect (the payment call that may or may not have landed) resolved?
4. Can derived views be deleted and rebuilt from durable history?
5. Which named identity submitted the work, and does every effect trace back to it?
6. Where does a policy decision meet the execution path, and does the record connect the decision to the resulting action?
7. Can an agent widen its own authority without an external decision?

8. How does a composition change take effect for work already in progress, and can the system say which exact versions a turn used?
9. Can the same definition run locally and in managed placement, and what cannot move?
10. Which system owns desired state, which owns executed state, and can an evidence product point back to the source record?

Appendix: glossary

Friendly term	Runtime meaning
agent definition	versioned declaration that begins a thread with an opening set of attachments
tool	model-visible surface of an executable operation
operation	immutable executable contract with typed input, output, effects, and requirements
install or enable	attach a published operation to a scope under policy
disable or revoke	detach the operation or narrow its attachment
thread	durable identity and ordered history of one scope of work
turn	one submitted continuation cycle ending at quiescence or budget
context	deterministic assembly of selected facts and views for a model call, with a receipt
memory	product term for durable facts, produced summaries, retrieval views, and context sources
workflow	strict policy over operations and agents
agent	adaptive policy connected to a durable thread and its attachments; the propagator on that thread

Friendly term	Runtime meaning
coupling	any function acting on a thread, with declared reads, writes, and trigger
propagator	coupling that writes back into the stream it reads; the agent loop
projection	coupling that derives output elsewhere (summary, index); lossy, with provenance
controller	coupling that writes into a control stream; what products call a hook
chaotic	a coupling whose output cannot be recomputed from its inputs; model-backed by definition
subagent	delegated child thread with its own durable identity and outcome
principal	the human or service identity on whose authority work entered
receipt	durable explanation of what the runtime resolved or did
placement	selected execution backend and runtime instance
registry	catalog and distribution system for immutable definitions and aliases
evidence view	searchable or mapped representation derived from runtime facts

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Status note, 21 August 2026: DeepSeek Harness is a developer preview. Cordis is under active development. Verlet is experimental. Every runtime claim in this document was checked against the Verlet Kernel v0.4.0 tag on that date.